

XIAOZHEN WU

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🎓 EDUCATION

Wuhan University | Wuhan, China

Sep 2024 – Jun 2027

Master of Arts in Intelligent Communication

- **Selected Coursework:** Social Network Computation, Optimization Theory and Methods, Operations Research, Linear Algebra, Advanced Mathematics, Modern AI Algorithms.
- **Honors:** First-class Graduate Academic Scholarship (2025), National 1st Prize in the 22nd "Huawei Cup" Chinese Graduate Mathematical Modeling Competition.

Wuhan University | Wuhan, China

Sep 2020 – Jun 2024

Bachelor of Arts in Network Communication

- **GPA:** 3.92/4.00 (Major Rank: 1/42)
- **Honors:** "Outstanding Student" Title (2020–2023), 14th Blue Bridge Cup Web Application Development National Excellence Award.

📄 PUBLICATIONS

- Zhang, M., Dong, Q., & **Wu, X.** *How misinformation diffuses on online social networks: radical opinions, adaptive relationship and algorithmic intervention.* IEEE Transactions on Computational Social Systems (SCI Q1).
 - Designed dynamic models to capture the co-evolution of users' misinformation beliefs and social relations under algorithmic interventions.
 - Simulated the impact of five content distribution mechanisms on social network structures.
- Dong, Q., Zhang, M., Wu, S., & **Wu, X.** *EMIF: Evidence-aware Multi-source Information Fusion Network for Explainable Fake News Detection.*
 - Proposed a co-attention network to capture semantic conflicts between comments and news articles, enhancing cross-source evidence consistency.

🏢 RESEARCH EXPERIENCE

Platform Swinging Dynamics & Opinion Evolution Modeling

Current

Independent Researcher Wuhan University

- Developing a data-driven simulation framework utilizing a GNN-LLM architecture to model cross-platform discourse shifts.
- Integrating Natural Language Processing with multi-agent systems to parse semantic nuances, using the competitive table tennis community as a primary cross-platform dataset.

High-Speed Train Bearing Cross-Domain Fault Diagnosis

Sep 2025

Core Developer Academic Project

- Designed a diagnostic framework combining deep transfer learning and multi-dimensional explainability using ResNet-18 (achieving F1 > 99.9% on source domain).
- Implemented Unsupervised Domain Adaptation (UDA) utilizing Domain Adversarial Neural Networks (DANN) with high-confidence pseudo-labels to solve cross-domain alignment challenges.

Mechanism of Misinformation Diffusion in Multilayer Social Networks

Jul 2022 – Aug 2023

Core Team Member Provincial Innovation Project

- Investigated the diffusion mechanisms of misinformation across platforms, conceptualizing multi-layer coupled network corresponding relationships.

- Constructed simulated scale-free and community coupled networks using NetLogo, and conducted simulation experiments via the BehaviorSpace module.

PROFESSIONAL EXPERIENCE

iFlytek Healthcare Co., Ltd. | Internship

May 2025 – Sep 2025

Assistant Research Algorithm Engineer

- **Epidemiological Simulation & Complex Dynamics:** Developed an Agent-Based Modeling (ABM) system and 7 compartmental models (SIR/MSEIR) to quantify policy interventions. Achieved precise epidemic trend simulation by restructuring ODEs and dynamically calculating R_0 .
- **NER System Optimization:** Reconstructed the address recognition module using a Bi-LSTM-CRF architecture for automated medical geographic information extraction. Resolved ambiguity in complex texts via Badcase attribution and incremental training.
- **Multi-Task Learning (MTL) Joint Model:** Built an MTL framework for simultaneous food type recognition and contaminated diet detection. Improved F1-score from 76% to 94% by using LLM-assisted generation for 6000+ samples and introducing Focal Loss.
- **Spatiotemporal Analysis & Prediction:** Implemented a Teacher-Student heterogeneous network using KL divergence and Laplacian graph regularization for cross-domain knowledge transfer. Optimized NxN matrix deduplication to accelerate scanning speeds for massive case data.

MiraclePlus (YC China) | Internship

Feb 2023 – Jun 2023

Product Intern

- **CRM Data Integration:** Collaborated with the development team to design matching rules for CRM systems, implementing fuzzy matching for key fields to ensure data integration accuracy.
- **Data Tracking & Analytics:** Constructed cross-channel data tracking systems by refactoring UTM parameter transmission logic and writing SQL queries in Superset to build conversion funnel dashboards.
- **Data Structuring via AI:** Participated in Prompt optimization strategies, utilizing ChatGPT's generative capabilities to extract and structure non-standardized information from application forms.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, Java, JavaScript
- **Frameworks & Modeling:** PyTorch, TensorFlow, NetLogo, GNN, LLM Integration, SQL